Task

Consider the Lotka-Volterra model with prey logistic growth:

1. Find the system's equilibrium points.
2. Calculate the Jacobian matrices of the system at the equilibrium points.
3. Calculate the eigenvalues of the Jacobian matrices.
4. Discuss the results.

Solution

Consider the following system of differential equations:

$$\begin{cases} \dfrac{dx}{dt} = rx\left(1 - \dfrac{x}{K}\right) - axy \\ \dfrac{dy}{dt} = bxy - dy \end{cases} \tag{1}$$

where

- x: Prey population
- y: Predator population
- r: Prey growth rate
- K: Prey environment carrying capacity
- a: Predation rate
- b: Predator reproduction rate
- d: Predator death rate

1. Finding the equilibrium points

$$\begin{cases} \dfrac{dx}{dt} = 0 \\ \dfrac{dy}{dt} = 0 \end{cases} \Rightarrow \begin{cases} rx\left(1 - \dfrac{x}{K}\right) - axy = 0 \\ bxy - dy = 0 \end{cases} \Rightarrow \begin{cases} x\left(r\left(1 - \dfrac{x}{K}\right) - ay\right) = 0 \\ y\left(bx - d\right) = 0 \end{cases} \tag{2}$$

Therefore, one of the equilibrium points is $(x = 0, y = 0) \rightarrow$ total extinction.

Now to analyze the case $(x \neq 0, y = 0)$:

$$\begin{aligned} &r\left(1 - \frac{x}{K}\right) - ay = 0 \quad \xRightarrow{y=0} \quad r - \frac{rx}{K} = 0 \Rightarrow \\ &\Rightarrow \frac{rx}{K} = r \quad \Rightarrow \quad x = K \end{aligned} \tag{3}$$

Another equilibrium point is $(x = K, y = 0) \to$ predator extinction, prey reaches carrying capacity.

Last case, $(x \neq 0, y \neq 0)$:

$$\begin{cases} r\left(1 - \frac{x}{K}\right) - ay = 0 \\ bx - d = 0 \end{cases} \tag{4}$$

$$bx - d = 0 \quad \Rightarrow \quad x = \frac{d}{b} \tag{5}$$

By substituting x in the first equation we obtain:

$$r\left(1 - \frac{d}{bK}\right) - ay = 0 \quad \Rightarrow \quad y = \frac{r(bK - d)}{abK} \tag{6}$$

The final equilibrium point is $(x = \frac{d}{b}, y = \frac{r(bK-d)}{abK}) \to$ coexistence.

2. Calculating the Jacobian matrix

Let $\frac{dx}{dt} = f(x, y)$ and $\frac{dy}{dt} = g(x, y)$. Then the associated Jacobian is:

$$J = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{bmatrix} \tag{7}$$

By calculating the partial derivatives we obtain:

$$J = \begin{bmatrix} r - \frac{2rx}{K} - ay & -ax \\ by & bx - d \end{bmatrix} \tag{8}$$

3. Evaluating the Jacobian at the equilibrium points

Case 1: $(x = 0, y = 0)$

$$J = \begin{bmatrix} r & 0 \\ 0 & -d \end{bmatrix} \tag{9}$$

Case 2: $(x = K, y = 0)$

$$J = \begin{bmatrix} -r & -aK \\ 0 & bK - d \end{bmatrix} \tag{10}$$

Case 3: $(x = \frac{d}{b}, y = \frac{r(bK-d)}{abK})$

$$J = \begin{bmatrix} -\frac{rd}{bK} & -\frac{ad}{b} \\ \frac{r(bK-rd)}{aK} & 0 \end{bmatrix} \tag{11}$$

3. Calculating eigenvalues

General formula:

$$\det(J - \lambda I) = 0 \tag{12}$$

For 2×2 matrices:

$$\lambda^2 - \lambda \operatorname{Tr}(J) + \det(J) = 0 \Leftrightarrow \tag{13}$$

$$\Leftrightarrow \lambda_{1,2} = \frac{\operatorname{Tr}(J) \pm \sqrt{\operatorname{Tr}(J)^2 - 4\det(J)}}{2} \tag{14}$$

Case 1

$$\left.\begin{matrix} \operatorname{Tr}(J) = r - d \\ \det(J) = -rd \end{matrix}\right\} \Rightarrow \lambda^2 - \lambda(r-d) - rd = 0 \Rightarrow \tag{15}$$

$$\Rightarrow (\lambda - r)(\lambda + d) = 0 \Rightarrow$$

$$\Rightarrow \begin{cases} \lambda_1 = r \\ \lambda_2 = -d \end{cases} \tag{16}$$

Case 2

$$\det(J - \lambda I) = \begin{vmatrix} -r - \lambda & -aK \\ 0 & bK - d - \lambda \end{vmatrix} = (-r - \lambda)(bK - d - \lambda) = 0 \Rightarrow$$

$$\Rightarrow \begin{cases} -r - \lambda = 0 \\ bK - d - \lambda = 0 \end{cases} \Rightarrow \begin{cases} \lambda_1 = -r \\ \lambda_2 = bK - d \end{cases} \tag{17}$$

Case 3
To be solved later...